HR Analytics Dashboard Report

Project Title: Employee Attrition Analysis
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Tools Used: Python (Pandas), Power BI, Microsoft Excel
Dataset Used: HR Employee Attrition Dataset
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Introduction

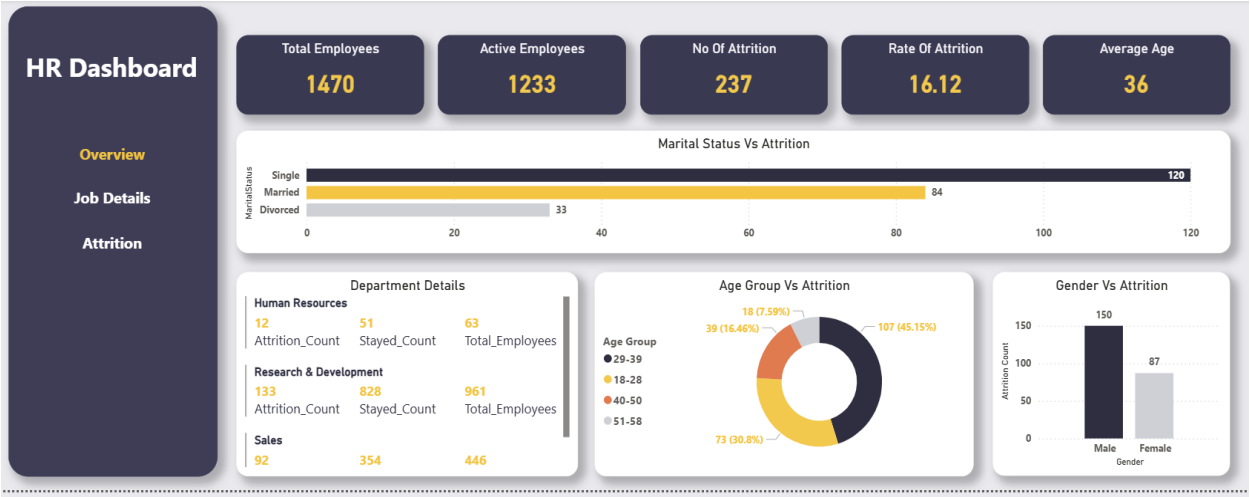
The objective of this project is to analyze employee attrition patterns and identify key factors contributing to workforce turnover. The dashboard provides insights into demographic, job-related, and compensation-based factors affecting employee retention.

This analysis helps organizations make data-driven HR decisions to improve employee engagement and reduce attrition costs.

About Dataset

Uncover the factors that lead to employee attrition and explore important questions such as ‘show me a breakdown of distance from home by job role and attrition’ or ‘compare average monthly income by education and attrition’. This is a fictional data set created by IBM data scientists.

Overview Dashboard Page



Visuals Included:

- Total Employees
- Active Employees
- Number of Attrition
- Attrition Rate (%)
- Average Age
- Marital Status vs Attrition
- Department Details
- Age Group vs Attrition
- Gender vs Attrition

Key Insights:

The analysis reveals clear attrition patterns across demographic and departmental factors.

- Single employees show the highest attrition, followed by married employees, while divorced employees have the lowest turnover. Department-wise breakdown highlights variations in total employees, active employees, and attrition across departments.
- The **29–39 age group** accounts for the highest attrition at **45%**, followed by **18–28 (30%)**, **40–50 (16%)**, and **51–58 (7%)**, indicating that mid-career professionals are more likely to leave.
- Gender analysis shows higher attrition among males, with **150 males** employees leaving compared to **87 females.**

Job Details Page

Visuals Included:

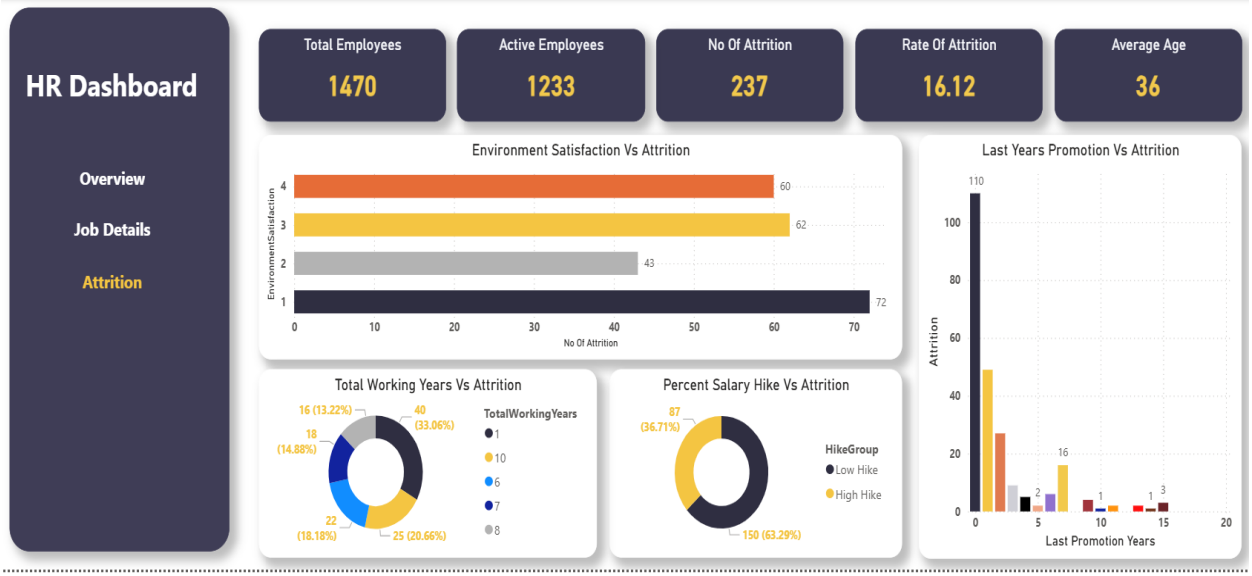
- Job Level vs Attrition
- Job Satisfaction Detail
- Job Role vs Attrition
- Education Field vs Attrition

Key Insights:

- Attrition is highest among Job Level 1 employees (143), followed by Level 2 (52) and Level 3 (32), while Level 4 and 5 show very low attrition (5 each). This indicates that junior-level employees are more likely to leave the organization.
- Role-wise analysis shows the highest attrition among Laboratory Technicians (62) and Sales Executives (57), followed by Research Scientists (47) and Sales Representatives (33). Other roles report comparatively lower turnover.
- Education field analysis reveals that employees from Life Sciences (89) and Medical (63) backgrounds contribute the most to attrition, followed by Marketing (35) and other fields.
- Overall, attrition is concentrated among lower job levels, specific technical and sales roles, and certain educational backgrounds.

This page focuses on role-based and satisfaction-driven analysis.

Promotion, Salary & Attrition Page



Visuals Included:

- Environment Satisfaction vs Attrition
- Years Since Last Promotion vs Attrition
- Total Working Years vs Attrition
- Percent Salary Hike vs Attrition

Key Insights:

- Employees with **Environment Satisfaction Level 1 (72)** show the highest attrition, indicating that lower workplace satisfaction contributes to turnover.
- The highest attrition is observed among employees with **1 year of total working experience (40 employees)**, suggesting early-career employees are more likely to leave.
- Salary analysis shows that employees with a **Low Salary Hike (150)** have higher attrition compared to those with a **High Salary Hike (87)**.
- Additionally, employees who were **promoted in the current year (110)** also show a high attrition count.

Overall, attrition is influenced by work environment satisfaction, limited experience, and lower salary growth.

Overall Findings

- **Research & Development** department shows relatively higher attrition compared to other departments.
- Employees with low job satisfaction and limited salary growth are more likely to leave.
- Early-career employees contribute significantly to overall attrition.
- The **29–39** age group accounts for the highest attrition at **45%.**
- Role-wise analysis shows the highest attrition among **Laboratory Technicians 62** which is 62% of overall.

Conclusion

The HR Analytics Dashboard successfully identifies major drivers of employee attrition.

Key focus areas for management:

- Improve career growth opportunities
- Enhance job satisfaction
- Review compensation structure
- Monitor high-risk departments

By implementing data-driven retention strategies, organizations can reduce turnover costs and improve workforce stability.

End of Report